

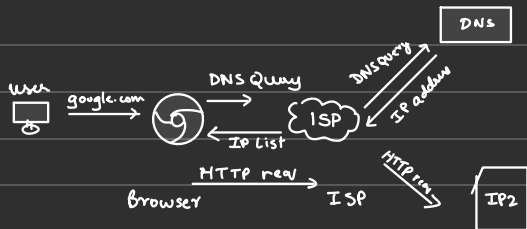
7 DNS : Domain Name Server

What is DNS?

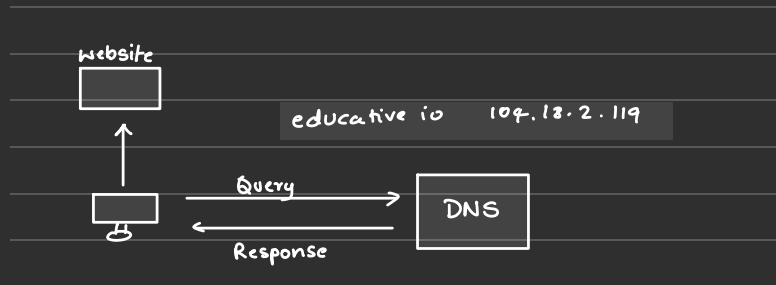
The **domain name system (DNS)** is the Internet's naming service that maps human-friendly domain names to machine-readable IP addresses. The service of DNS is transparent to users. When a user enters a domain name in the browser, the browser has to translate the domain name to IP address by asking the DNS infrastructure. Once the desired IP address is obtained, the user's request is forwarded to the destination web server.

The slides below show the high-level flow of the working of DNS:

- DNS ? Naming service of Internet
- maps human friendly domain name to IP Address



The entire operation is performed very quickly. Therefore, the end user experiences minimum delay. We'll also see how browsers save some of the frequently used mappings for later use in the next lesson.



Important details

Let's highlight some of the important details about DNS, some of which we'll cover in the next lesson:

Name servers: It's important to understand that the DNS isn't a single server. It's a complete infrastructure with numerous servers. DNS servers that respond to users' queries are called **name servers**.

Resource records: The DNS database stores domain name to IP address mappings in the form of resource records (RR). The RR is the smallest unit of information that users request from the name servers. There are different types of RRs. The table below describes common RRs. The three important pieces of information are *type*, *name*, and *value*. The *name* and *value* change depending upon the *type* of the RR.

Resource Record Types:

Type	Description	Name	Value	eg
book.				
A	Provides host name to IP address mappings	Host name	IP address	(A, edu.io, 104.18.2.119)
NS	Provides HN i.e authoritative DNS for a Domain name	Domain Name	Host name	(NS, edu.io, dns.edu.io)
CNAME	mapping from alias to canonical hostname	HN	canonical name	(CNAME, edu.io, server1.primary.edu.io)
MX	mapping of mail server from alias to Canonical HN	HN	canonical name	(mx, mail.edu.io, mailserver1.backup.edu.io)

- Caching : reduces latency.

Caching: DNS uses caching at different layers to reduce request latency for the user. Caching plays an important role in reducing the burden on DNS infrastructure because it has to cater to the queries of the entire Internet.

Hierarchy: DNS name servers are in a hierarchical form. The hierarchical structure allows DNS to be highly scalable because of its increasing size and query load. In the next lesson, we'll look at how a tree-like structure is used to manage the entire DNS database.

Let's explore more details of the above points in the next lesson to get more clarity.

How the Domain Name System Works

Through this lesson, we'll answer the following questions:

How is the DNS hierarchy formed using various types of DNS name servers?

How is caching performed at different levels of the Internet to reduce the querying burden over the DNS infrastructure?

How does the distributed nature of the DNS infrastructure help its robustness?

DNS hierarchy

As stated before, the DNS isn't a single server that accepts requests and responds to user queries. It's a complete infrastructure with name servers at different hierarchies.

There are mainly four types of servers in the DNS hierarchy:

DNS resolver: Resolvers initiate the querying sequence and forward requests to the other DNS name servers. Typically, DNS resolvers lie within the premise of the user's network. However, DNS resolvers can also cater to users' DNS queries through caching techniques, as we will see shortly. These servers can also be called local or default servers.

Root-level name servers: These servers receive requests from local servers. Root name servers maintain name servers based on top-level domain names, such as .com, .edu, .us, and so on. For instance, when a user requests the IP address of educative.io, root-level name servers will return a list of top-level domain (TLD) servers that hold the IP addresses of the .io domain.

Top-level domain (TLD) name servers: These servers hold the IP addresses of authoritative name servers. The querying party will get a list of IP addresses that belong to the authoritative servers of the organization.

Authoritative name servers: These are the organization's DNS name servers that provide the IP addresses of the web or application servers.

Hierarchy : Not a single server. 4 types:

1. **DNS Resolver** : local / default servers.

Initiates DNS query sequence.

Resides within user's network.

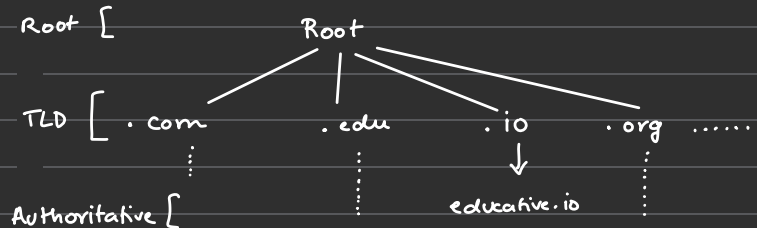
User caching : reduces response time

2. **Root level name servers** : .com, .edu, .us etc.

Top level domain (TLD) IP address of .io

3. **TLD servers** : IP address of authoritative servers of servers.

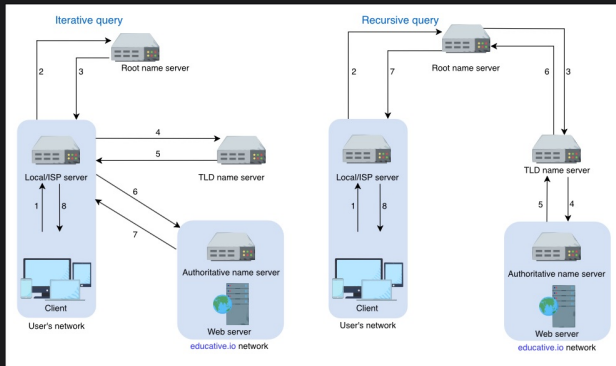
4. **Authoritative servers** : IP address of web app server



There are two ways to perform a DNS query:

Iterative: The local server requests the root, TLD, and the authoritative servers for the IP address.

Recursive: The end user requests the local server. The local server further requests the root DNS name servers. The root name servers forward the requests to other name servers.



Iterative versus recursive query

Caching

Caching refers to the temporary storage of frequently requested resource records. A **record** is a data unit within the DNS database that shows a name-to-value binding. Caching reduces response time to the user and decreases network traffic. When we use caching at different hierarchies, it can reduce a lot of querying burden on the DNS infrastructure. Caching can be implemented in the browser, operating systems, local name server within the user's network, or the ISP's DNS resolvers.

The slideshow below demonstrates the power of caching in the DNS:

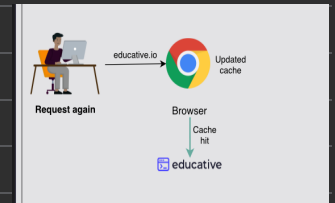
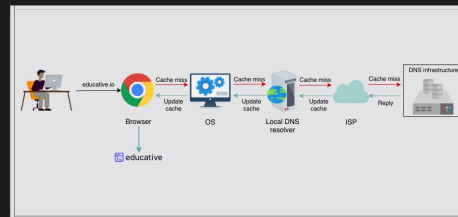
The user requests to visit a URL, and the browser has cached the domain name to IP address mapping

Here's what happens if the browser hasn't cached the domain to IP address mapping. The next hierarchy that can have the mapping is the OS

If the OS doesn't have the mapping, the local DNS resolver can have the cached response

If the local DNS resolver doesn't have the mapping, the ISP can have a cached response

The cache will be updated at each hierarchy



Although the DNS hierarchy facilitates the distributed Internet that we know today, it's a distributed system itself. The distributed nature of DNS has the following advantages:

It avoids becoming a single point of failure (SPOF).

It achieves low query latency so users can get responses from nearby servers.

It gets a higher degree of flexibility during maintenance and updates or upgrades. For example, if one DNS server is down or overburdened, another DNS server can respond to user queries.

- **Highly Scalable:** 1000 replicas of 13 root level servers spread thru the world

- **Reliable:**

- i) caching: browser, OS, local server, ISP DNS

- ii) Server replicas: global DNS

- iii) Protocol: UDP - faster $\Rightarrow$ performance

- **Consistency?** Not STRONG. More Reading? WAITING

Cache: TTL - time to live. update outdated caches

DNS as a distributed system

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There are 13 logical root name servers (named letter **A** through **M**) with many instances spread throughout the globe. These servers are managed by 12 different organisation

Highly scalable

Due to its hierarchical nature, DNS is a highly scalable system. Roughly 1,000 replicated instances of 13 root-level servers are spread throughout the world strategically to handle user queries. The working labor is divided among TLD and root servers to handle a query and, finally, the authoritative servers that are managed by the organizations themselves to make the entire system work. As shown in the DNS hierarchy tree above, different services handle different portions of the tree enabling scalability and manageability of the system.

Three main reasons make the DNS a reliable system:

Caching: The caching is done in the browser, the operating system, and the local name server, and the ISP DNS resolvers also maintain a rich cache of frequently visited services. Even if some DNS servers are temporarily down, cached records can be served to make DNS a reliable system.

Server replication: DNS has replicated copies of each logical server spread systematically across the globe to entertain user requests at low latency. The redundant servers improve the reliability of the overall system.

Protocol: Although many clients rely on the unreliable User Datagram Protocol (UDP) to request and receive DNS responses, it's important to acknowledge that UDP also offers distinct advantages. It is much faster, and therefore, improves DNS performance. Furthermore, internet service reliability has improved since its inception, so UDP is usually favored over TCP. DNS queries are usually retransmitted at the transport layer if there's no response for the previous one. Therefore, request-response might need additional round trips, which provides a shorter delay as compared to TCP, which needs a three-way handshake every time before data exchange.

Consistent

DNS uses various protocols to update and transfer information among replicated servers in a hierarchy. DNS compromises on strong consistency to achieve high performance because data is read frequently from DNS databases as compared to writing. However, DNS provides eventual consistency and updates records on replicated servers lazily. Typically, it can take from a few seconds up to three days to update records on the DNS servers across the Internet. The time it takes to propagate information among different DNS clusters depends on the DNS infrastructure, the size of the update, and which part of the DNS tree is being updated.

Consistency can suffer because of caching too. Since authoritative servers are located within the organization, it may be possible that certain resource records are updated on the authoritative servers in case of server failures at the organization. Therefore, cached records at the default/local and ISP servers may be outdated. To mitigate this issue, each cached record comes with an expiration time called **time-to-live (TTL)**.